

Practice Exam Version # 2

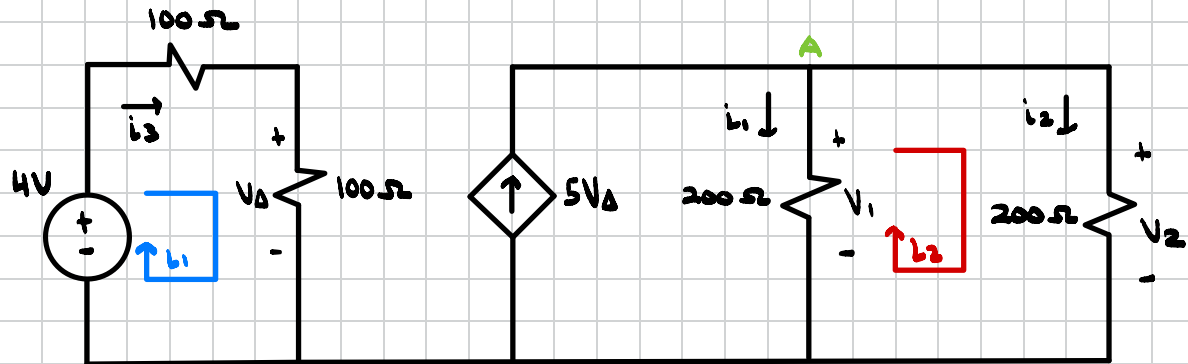
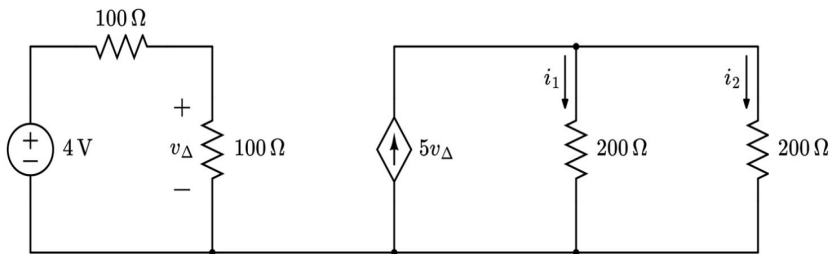
Question One: KCL, KVL, Ohm's Law, and Passive Sign Convention

For the circuit shown below, find the following:

- a) i_1
- b) i_2

Finally, calculate the power of the furthest right $200\ \Omega$ resistor and state whether the power is absorbed or delivered.

Suggested time limit: 20 minutes



Step #1 KVL @ L_1

$$-4 + 100i_3 + 100i_3 = 0$$

$$-4 + 200i_3 = 0$$

$$200i_3 = 4$$

$$i_3 = 20\text{mA}$$

Step #2 Find V_A

$$V_A = IR \text{ Where } R = 100\Omega \text{ and } I = i_3$$

$$V_A = (20 \cdot 10^{-3})(100)$$

$$V_A = 2\text{V}$$

Step #3 KVL @ L_2

$$-V_1 + V_2 = 0$$

$$V_1 = V_2$$

$$200i_1 = 200i_2$$

$$i_1 = i_2$$

Step #4 KCL @ Node A

$$-5V_A + i_1 + i_2 = 0$$

$$-5(2) + i_1 + i_1 = 0$$

$$-10 + 2i_1 = 0$$

$$2i_1 = 10$$

$$i_1 = i_2 = 5\text{A}$$

Step #5 Power of furthest 200Ω

$$P = I^2 R \text{ Where } I = i_2 \text{ and } R = 200\Omega$$

$$P = (5)^2(200)$$

$$P = 5000\text{W Absorb}$$

Grading Rubric

Find V_0 - 3 Points

Find i_1 = 2 Points

Find i_2 = 2 Points

Power of furthest 200 Ω - 2 Points

State Delivered or Absorb - 1 Point

Discussion #1 Problems To Practice

Problem # 5

Problem # 8

Problem # 9

Problem # 10

Problem # 12

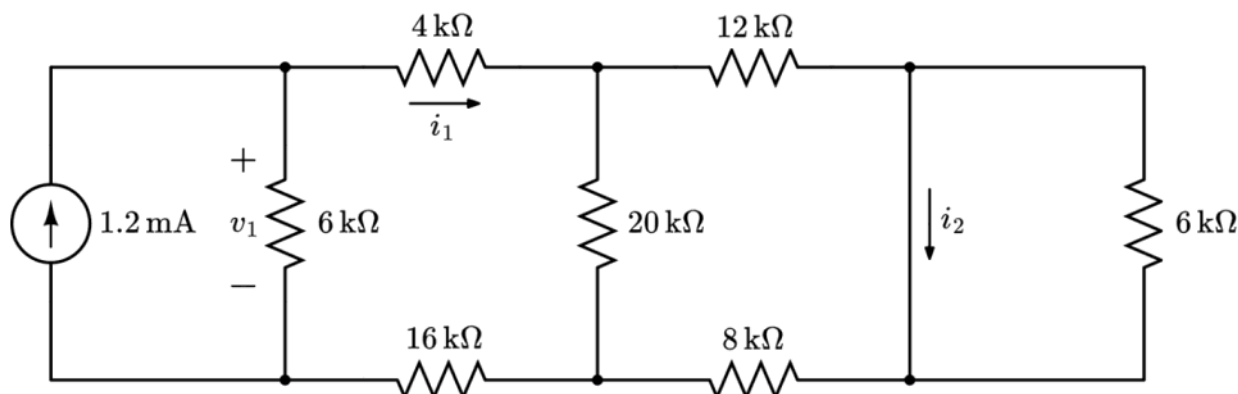
Problem # 13

Question Two: Current Division, Voltage Division, and Equivalent Resistance

For the circuit shown below, find the following:

- a) v_1
- b) i_1
- c) i_2

Step One : Equivalent Resistance seen by the Independent Source

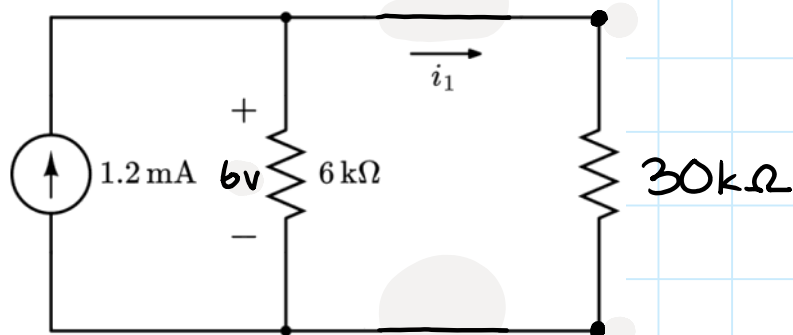


- ① $6k\Omega$ shorts
- ② $12k\Omega + 8k\Omega = 20k\Omega$
- ③ $20k\Omega \parallel 20k\Omega = 10k\Omega$
- ④ $10k\Omega + 4k\Omega + 16k\Omega = 30k\Omega$
- ⑤ $30k\Omega \parallel 6k\Omega = 5k\Omega$

Ohm's Law: $V_1 = 1.2mA \cdot 5k\Omega$ $V_1 = 6V$

Step Two: Find i_1 using Current Division

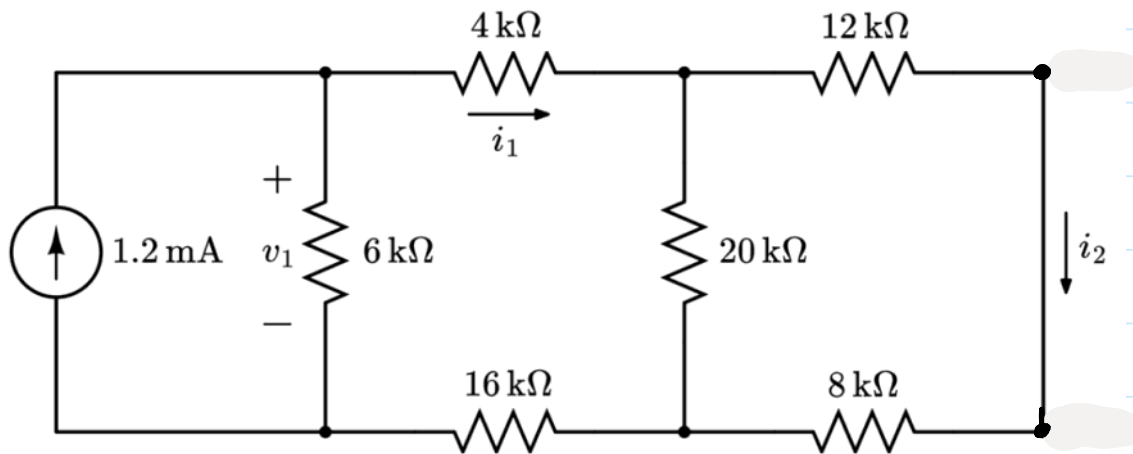
Sketching the circuit from equivalent resistance step 3, where i_1 flows through one branch



$$i_1 = 1.2mA \cdot \frac{5k\Omega}{30k\Omega}$$

$$i_1 = 0.2mA$$

Step Three: Find i_2 using current division



$$i_2 = 0.1 \text{ mA} \cdot \frac{10 \text{ k}\Omega}{20 \text{ k}\Omega}$$

$$i_2 = 0.1 \text{ mA}$$

The formula for current division contains equivalent resistance. Since we know i_1 , we are only finding the equivalent resistance for the branches after.

Self-Grading Rubric	
Identify the shorted resistor	+1 point
Correct equivalent resistance seen by source	+3 points
Identify $v_1 = 6 \text{ V}$	+2 points
Perform current division to find $i_1 = 0.2 \text{ mA}$	+2 points
Perform current division to find $i_2 = 0.1 \text{ mA}$	+2 point

Suggested discussion problems:

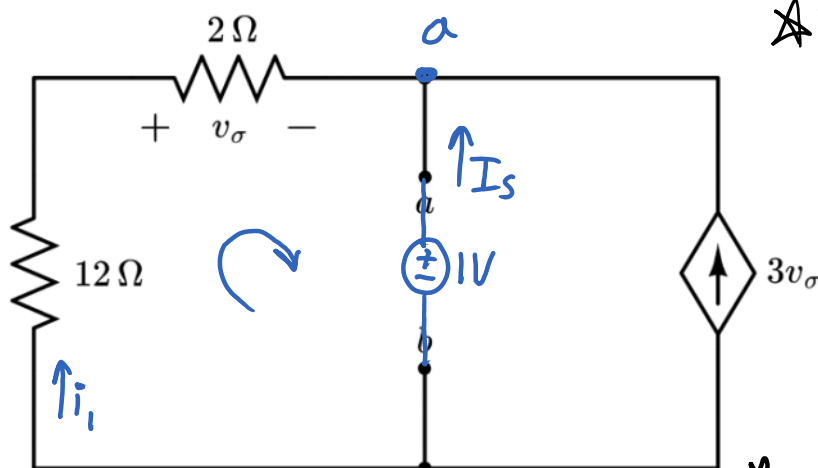
- Discussion 2 Questions 7-8
- Discussion 3 Questions 8-10

Practice Exam 1 Version 2 Q3 Solution

Question Three: Equivalent Resistance with Dependent Sources

For the circuit shown below, find the equivalent resistance seen by the open terminal pair ab .

Suggested time limit: 20 minutes



★ To find R_{eq} with a dependent source:

$$R_{eq} = \frac{V_s}{I_s}$$

Find i_1 :

KVL left loop:

$$12i_1 + 2i_1 + 1 = 0$$

$$i_1 = -\frac{1}{14} \text{ A}$$

I defined i_1 to follow passive sign convention.

Find V_0 :

$$V_0 = 2i_1 = 2\left(-\frac{1}{14}\right) = -\frac{1}{7} \text{ V}$$

KCL @ a :

$$-i_1 - I_s - 3V_0 = 0$$

$$-\left(-\frac{1}{14}\right) - I_s - 3\left(-\frac{1}{7}\right) = 0$$

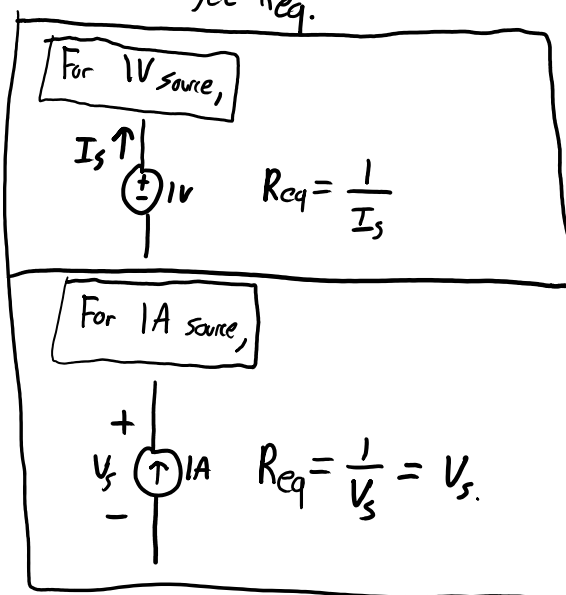
$$\frac{1}{14} - I_s + \frac{3}{7} = 0$$

$$I_s = \frac{1}{2} \text{ A}$$

$$R_{eq} = \frac{1}{I_s} = \frac{1}{\frac{1}{2}} = 2 \Omega$$

$$R_{eq} = 2 \Omega$$

★ If we give V_s or I_s an arbitrary value (such as 1V or 1A), we can solve for the other value to get R_{eq} .



★ You can choose any way you want! I think a 1V source is simpler for this specific problem, so I'll use that.

Grading Rubric

Find i_1 : +3 Points

Find V_o : +2 Points

Find I_s : +3 Points

Find R_{eq} : +2 Points

Discussion 3 Problems To Practice:

- Problem (4)

- Problem (5)

- Problem (6)

- Problem (7)